

Bannaga Altieb Abdul Muhsin

Electrical Project Engineer — Riyadh, Saudi Arabia

Telecom & industrial passive infrastructure · Energy optimization · O&M excellence

CEM® · CMVP® · LEED Green Associate

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PROFESSIONAL SUMMARY

Certified Energy Manager (CEM®) and Certified Measurement & Verification Professional (CMVP®) with 14+ years of experience in electrical and energy engineering. Specializes in project management, telecom power systems, energy-efficient solutions, and technical operations across diverse infrastructure. Expertise spans data collection, energy analysis, facilities management, AI-driven dashboards, and structured reporting. Delivered innovative, reliable, and performance-driven engineering solutions across telecom, power generation, and industrial sectors. Holds a B.Sc. in Electronic Control Engineering with additional certifications in LEED Green Associate and SCADA systems. Fluent in English and Arabic.

OVERVIEW

14+ Years Experience	15+ Certifications & Training	60+ MV/LV Assets Commissioned	18% Utility Cost Reduction
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AI & DEVELOPMENT

AI & Automation

- Designed Python-based automation to streamline energy performance reporting and KPI tracking across sites.
- Developed machine-learning-based anomaly detection alerts for telecom site power consumption behavior.
- Built governance-ready Energy & Environmental dashboards for real-time CO2 footprint monitoring.
- Applied Power BI, SQL Server, and DAX to model and visualize operational trends across multi-site networks.

Development & Tools

- Created reusable troubleshooting guides and diagnostic scripts for SCADA / PLC / DCS systems.
- Documented API-style checklists to standardize field data capture and site integrations.
- Version-controlled documentation and engineering templates using Git/GitHub workflows.

KEY ACCOMPLISHMENTS

- Reduced nationwide telecom site utility costs by up to **18%** through multi-phase energy optimization programs.
- Delivered uninterrupted O&M coverage for STC COW & Mobily ENM with proactive outage prevention strategies.
- Authored troubleshooting guides and automation routines reducing incident resolution time by **30%**.
- Commissioned **60+ MV/LV assets** under SEC unified contracts with full SCADA integration and zero-defect handover.
- Developed an AI-driven Energy & Environmental Dashboard tracking real-time CO2 emissions and power efficiency.
- Applied Power BI, Python, and SQL Server to model operational trends across 600+ multi-site telecom networks.
- Version-controlled all engineering documentation and templates using Git/GitHub workflows.

EXPERIENCE

Electrical Project Engineer

2019 – Present

ACES | Riyadh, Saudi Arabia

- Serve as Energy Manager and efficiency designer across STC COW and Mobily ENM managed networks.
- O&M Engineer for Mobily ENM Managed Services — full facility operations, CAFM/BMS integration, and KPI reporting.
- O&M Engineer for STC COW Managed Services — operations and maintenance across 600+ nationwide telecom sites.
- Designed and delivered multi-phase energy optimization programs achieving up to 18% reduction in utility costs.
- Developed Python-based automation for energy performance reporting and ML-based anomaly detection alerts.
- Built AI-driven Energy & Environmental Dashboard for CO2 footprint monitoring across all STC COW sites.
- Authored comprehensive technical troubleshooting and preventive maintenance documentation.
- Mentored junior engineers and standardized field engineering practices across all sites.

Electrical Project Engineer

2018 – 2019

Absar | Riyadh, Saudi Arabia

- Executed SEC unified-contract builds for 60+ MV/LV electrical assets including RMUs, transformers, and LV switchgear.
- Performed full commissioning, testing, and zero-defect SEC handover documentation.
- Installed and configured SCADA monitoring and control system integration.

Electrical Engineer

2017 – 2018

BEMCO

- Preventive and corrective maintenance for 11kV distribution networks.
- Developed standardized testing, inspection, and commissioning manuals.
- Managed NCR resolutions and maintained technical documentation workflows.

Electrical Engineer

2012 – 2017

SEC Thermal Generation | Sudan

- Day-to-day operations and maintenance for a 200 MW thermal generation block.
- Managed wastewater treatment systems using PH meters and OC 4000 DCS controls.
- Preventive maintenance and fault management for 11kV distribution systems.
- Developed operational procedures, safety documentation, and switching routines.

KEY PROJECTS

STC COW Network — O&M & Energy Optimization

End-to-end operations and maintenance for 600+ telecom sites nationwide. Delivered systematic energy auditing, SCADA oversight, and multi-phase optimization achieving up to 18% utility cost reduction.

Mobily ENM Managed Services

Full O&M and facility operations for Mobily's ENM managed services network including CAFM/BMS integration, outage prevention, and structured performance reporting.

Mobily Data Center Energy Audit

Comprehensive energy audit of critical data center infrastructure delivering measurable PUE improvements and quantified energy savings certified under CEM® / CMVP® methodology.

SEC Unified Contract — MV/LV Commissioning

Complete installation and commissioning of 60+ MV/LV electrical assets (RMUs, transformers, mini-pillars, LV switchgear) with full SCADA integration and zero-defect SEC handover documentation.

PP12 Thermal Generation Plant — 200 MW Operations

Day-to-day operations including 11kV switching, safety routines, wastewater treatment management, and DCS-based environmental monitoring for a 200 MW thermal power station.

AI CO2 & Energy Dashboard

Built ML-based anomaly detection and CO2 footprint tracking dashboards for nationwide STC COW telecom deployments using Python, Power BI, SQL Server, and DAX — enabling real-time environmental governance.

EDUCATION

B.Sc. Electronic Engineering (Control)
El Neelain University — Khartoum, Sudan

2007 – 2012

SKILLS

Power Systems & Electrical	HV/MV/LV Engineering · SCADA / DCS / PLC / RTU · Energy Auditing (CEM®) · Commissioning & Testing · Power System Protection · 11kV Switching · Transformer & RMU Installation
AI, Data & Analytics	Power BI / DAX · Python / Machine Learning · SQL Server · Data Dashboards · KPI Reporting · CO2 Monitoring · Anomaly Detection
Software & Development	React / Next.js · TypeScript · Tailwind CSS · Node.js · PostgreSQL · GraphQL · Git/GitHub
Design & Prototyping	Figma (96%) · Power BI Visuals (92%) · UI Systems (94%) · Dashboard Design · Wireframing
Energy & Facilities	Energy Management (CEM®/CMVP®) · CAFM · BMS Integration · Facility Operations · PUE Optimization · Preventive Maintenance · LEED Green Associate
Tools & Platforms	Microsoft Office Suite · Power BI · SQL Server · Figma · SCADA Platforms · Git/GitHub · Python

CERTIFICATIONS & TRAINING

CMVP®	Certified Measurement & Verification Professional
CEM®	Certified Energy Manager
ML	Machine Learning with Python
Power BI	Power BI Essential Training
LEED	LEED Green Associate (Cert Prep)
Excel	Learning Excel: Data Analysis
QE	Quality Engineering
SCADA	SCADA / PLC / DCS / RTU Systems
Data Sci	The Non-Technical Skills of Effective Data Scientists

MEMBERSHIPS

Saudi Council of Engineers (SCE)
Sudanese Engineers Association (SEA)

LANGUAGES

Arabic — Native
English — Professional

AVAILABILITY

Sun–Thu: 8:00 AM – 6:00 PM
Weekends: On-call for urgent needs
Open to short-notice regional travel

HOBBIES & INTERESTS

Electrical Power Systems · Energy Efficiency · Automation & Control · Data Analysis & Visualization · Innovation & New Technologies · AI & Machine Learning

"You never change things by fighting the existing reality. To change something, build a new model that makes the old one obsolete." — R. Buckminster Fuller